

Chapter

Electrolysis

1. An aqueous solution of copper sulphate turns colourless on electrolysis.

Which of the following could be the electrodes? SQP 2025

P. anode: copper; cathode: copper

Q. anode: platinum; cathode: copper

R. anode: copper; cathode: platinum

- (a) only P
- (b) only Q
- (c) only R
- (d) both Q and R

2. Which of the following electronic configuration represents the most electropositive element? SQP 2025

- (a) 2,1
- (b) 2,8,1
- (c) 2,2
- (d) 2,8,2

3. Identify one statement that holds true for electrolysis of molten lead bromide. COMP 2021

- (a) Silver grey metal deposits at the anode.
- (b) Temperature is not maintained during electrolysis.
- (c) Brown vapours of bromine are obtained at the anode.
- (d) Electrolyte contains H^+ ions along with Pb^{2+} ions.

4. Solid sodium chloride does not conduct electricity as. MAIN 2021

- (a) the strength of the bond is weak
- (b) it contains free ions
- (c) it does not contain any free ion
- (d) it contains free ions as well as molecules

5. Which statement about conduction of electricity is correct? SQP 2021

- (a) Electricity is conducted in aqueous solution by electrons.
- (b) Electricity is conducted in a metal wire by ions.
- (c) Electricity is conducted in a molten electrolyte by electrons.
- (d) Electricity is conducted in an acidic solution by ions.

6. Electrolysis of lead bromide is carried out in an : MAIN 2022

- (a) Aluminium vessel
- (b) Clay crucible
- (c) Silica crucible
- (d) Glass vessel

7. What is the difference between conductivity of Cu metal and of CuSO_4 solution ? COMP 2023

8. Compounds which conduct electricity in molten state. COMP 2023

9. Process of formation of ions from the molecules. SQP 2021

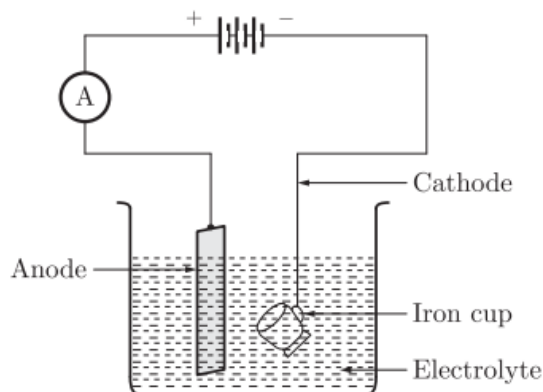
10. Process of separation of ions present in an ionic compound. MAIN 2022

11. (i) What should be the physical state of the lead bromide if it is to conduct electricity?

(ii) What particles are present in pure lead bromide?

(iii) Write the equations for the reactions which take place at electrodes during the electrolysis of lead bromide. SQP 2021

12. The following sketch represents the electroplating of an iron cup with Nickel metal. Study the diagram and answer the following questions.



- (a) During electroplating the iron cup is placed at the cathode. Why?
- (b) Name the ion that must be present in the electrolyte.
- (c) State one condition that is necessary to ensure that the deposit is smooth, firm and even.
- (d) Write the reaction taking place at the cathode.
- (e) What change would you observe at the anode?

13. Give the precautions to be taken during electroplating.

Solution

1. (b) only Q

If a **platinum anode** is used, it does not dissolve during electrolysis. Therefore, the **blue Cu^{2+} ions** removed from the solution at the cathode are **not replaced** at the anode. As a result, the concentration of Cu^{2+} ions decreases and the **blue colour of the CuSO_4 solution fades**, eventually turning colourless.

2. (b) 2,8,1

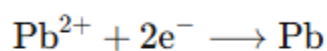
The electronic configuration **2,8,1** belongs to an alkali metal (like sodium). Elements with **one electron in the outermost shell** lose that electron very easily.

Hence, they are the **most electropositive**.

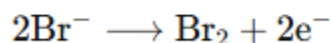
3. (c) Brown vapours of bromine are obtained at the anode

Electrolysis of molten PbBr_2 :

- **At the cathode (-):** Lead ions are reduced to lead metal



- **At the anode (+):** Bromide ions are oxidised to bromine gas



- The bromine gas appears as **brown vapours** at the anode.

4. (c) it does not contain any free ion

Explanation:

Solid sodium chloride consists of ions held **tightly in a crystal lattice**. Since the ions are **fixed in position**, they **cannot move freely** to conduct electricity.

5. (c) Electricity is conducted in a molten electrolyte by electrons.

- In a **molten electrolyte** (like molten sodium chloride), the ions are free to move.

- **Cations** move toward the cathode and **anions** move toward the anode, carrying the electric current.
- Thus, electricity in molten electrolytes is conducted by **ions**, not electrons in the metal sense.

6. (c) Silica crucible

- Electrolysis of **molten lead bromide (PbBr_2)** requires a vessel that **can withstand high temperatures** and **does not react with the electrolyte**.
- **Silica crucible** is **chemically inert** and **heat-resistant**, making it ideal for this purpose.

7.

Property	Copper (Cu) metal	Copper sulphate (CuSO_4) solution
Reason for conductivity	Presence of free electrons in the metal lattice	Presence of ions (Cu^{2+} and SO_4^{2-}) in solution
Type of charge carrier	Electrons	Ions
Nature of conduction	Metallic conduction	Electrolytic conduction

Explanation:

- In **copper metal**, **electrons are mobile** and carry the current.
- In **CuSO_4 solution**, **ions move** toward electrodes, carrying electricity through the solution.

8. Electrovalent compounds

9. Ionisation

10. Dissociation

11. (i) Physical state of lead bromide for conduction:

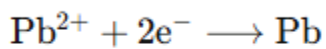
- Lead bromide must be **molten** (or dissolved in water) to conduct electricity.
- In solid state, ions are **fixed in the lattice** and cannot move.

(ii) Particles present in pure lead bromide:

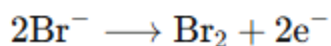
- Lead ions (Pb^{2+}) and bromide ions (Br^-)

(iii) Reactions at electrodes during electrolysis of lead bromide:

- **At the cathode (-):** Lead ions are reduced to lead metal



- **At the anode (+):** Bromide ions are oxidised to bromine gas



Thus, molten PbBr_2 conducts electricity and undergoes electrolysis with these electrode reactions.

12. (a) The metal which is to be electroplated should always be at cathode in electroplating. Therefore, the metal ion (Ni^{2+}) in it gets reduced at the cathode and forms a layer on it.

(b) Cation: Ni^{2+} and anion depend upon the electrolyte taken.

(c) The passage of low current (or long time is a necessary condition to ensure the deposit is smooth, even and firm.

(d) $\text{Ni}^{2+} + 2\text{e}^- \longrightarrow \text{Ni}$

(e) Negatively charge ions (depends upon the electrolyte) move toward the anode but do not discharged due to nature of the electrode. But Ni losses two electrons converted into Ni^{2+} in the solution. That results anode electrode get thinner and iron nip becomes thicker over the time of electrolysis.

13. Precautions to be taken during electroplating:

1. The **surface of the article** to be electroplated should be **smooth and uniform** to ensure a **firm and even deposit**.

2. The surface must be **clean and free from oil, grease, or dirt**.
3. The **emf of the battery** should not be too high to avoid rapid deposition.
4. A **smaller current** should be passed for a **longer time** to get a uniform layer.
5. The **electrolyte should not overheat**, because excessive heat can make the deposit **weak and uneven**.

These precautions ensure a **good quality electroplated layer**.